10.6 Circumference and arc length

EASY

Learning intention We are learning to find circumferences, arc lengths and perimeters of curved shapes.

Success criteria
- ☐ I can use C = πd or C = 2πr to find a circumference
- ☐ I can find an arc length: θ ÷ 360 × 2πr
- ☐ I can find the perimeter of sectors and composite shapes

NOTES — Key ideas for this lesson

TEACHER COPY

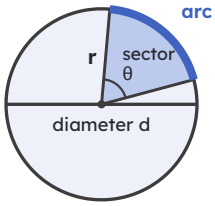
KEY TERMS

circumference (C): the distance around a circle
radius (r): from the centre to the edge
diameter (d): across through the centre; d = 2r
arc: part of the circumference
sector: a slice of a circle between two radii
θ: the angle at the centre of a sector
perimeter: the distance around the outside of a shape

FORMULAS

C = πd = 2πr
Arc length = $\frac{\theta}{360}$ × 2πr
Semicircle: P = ½πd + d
Quadrant: P = ¼ × 2πr + 2r
Sector: P = arc + 2r

STEPS 1 r = d ÷ 2 2 write the formula 3 substitute (π key) 4 round to 1 d.p. + unit



circumference

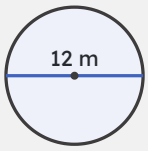
WE DO — With your teacher: copy what they write on the board.

Example 1 Find each circumference or perimeter, correct to one decimal place.

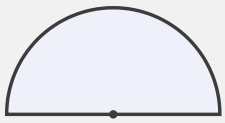
a Circumference



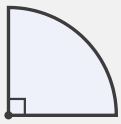
b Circumference



c Perimeter



d Perimeter



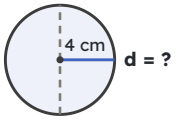
YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER d = 2r · r = d ÷ 2 · C = πd · Use the π key, then round to one decimal place.

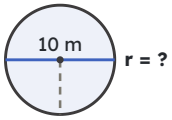
ROUND 1 — Questions 1–4: find the diameter or the radius.

1



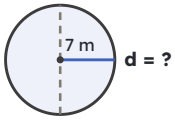
Formula: d = 2r
Substitute:
Answer:

2



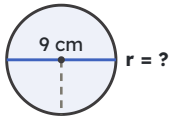
Formula: r = d ÷ 2
Substitute:
Answer:

3



Formula: d = 2r
Substitute:
Answer:

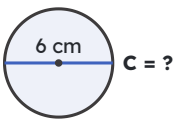
4



Formula: r = d ÷ 2
Substitute:
Answer:

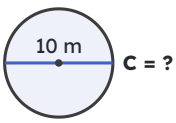
ROUND 2 — Questions 5–8: find the circumference. The diameter is given.

5



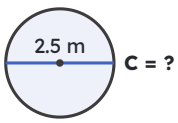
Formula: C = πd
Substitute:
Answer:

6



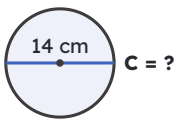
Formula: C = πd
Substitute:
Answer:

7



Formula: C = πd
Substitute:
Answer:

8



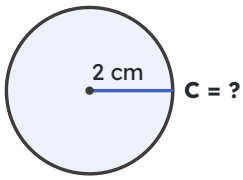
Formula: C = πd
Substitute:
Answer:

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER C = 2πr = πd · Semicircle P = ½πd + d · Quadrant P = ¼ × 2πr + 2r · Find d first when r is given.

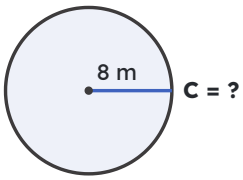
ROUND 3 Questions 9–11: find the circumference. The radius is given.

9



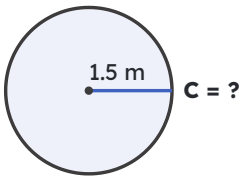
Formula C = 2πr
Substitute
Answer

10



Formula C = 2πr
Substitute
Answer

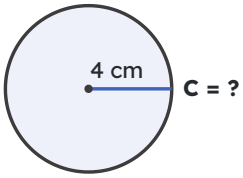
11



Formula C = 2πr
Substitute
Answer

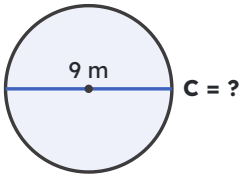
ROUND 4 Questions 12–14: is it r or d? Write the right formula, then find the circumference.

12



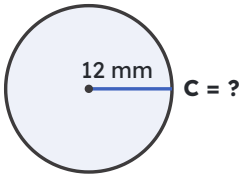
Formula
Substitute
Answer

13



Formula
Substitute
Answer

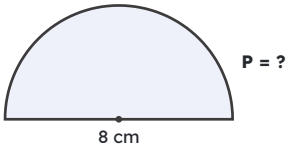
14



Formula
Substitute
Answer

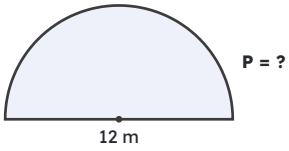
ROUND 5 Questions 15–17: find the perimeter of each semicircle.

15



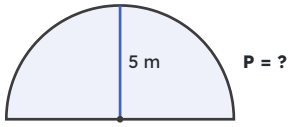
Formula P = ½πd + d
Substitute
Answer

16



Formula P = ½πd + d
Substitute
Answer

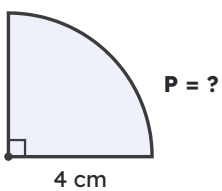
17



Formula P = ½πd + d
Substitute
Answer

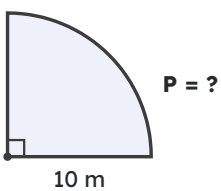
ROUND 6 Questions 18–20: find the perimeter of each quadrant.

18



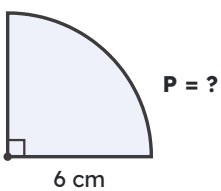
Formula P = ¼ × 2πr + 2r
Substitute
Answer

19



Formula P = ¼ × 2πr + 2r
Substitute
Answer

20



Formula P = ¼ × 2πr + 2r
Substitute
Answer

10.6 Circumference and arc length

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Give each answer correct to one decimal place.**

a Arc length

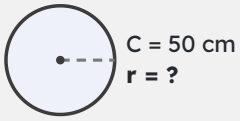


b Perimeter of the sector in part a

c Perimeter



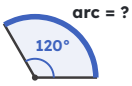
d Radius



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

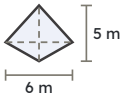
REMEMBER Arc = θ ÷ 360 × 2πr (θ = angle at the centre) · Sector P = arc + 2r · Composite: add the outside edges only

1 Arc length



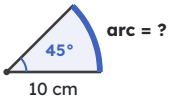
Formula _____
Substitute _____
Answer _____

2 Area 10.5



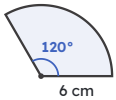
Formula _____
Substitute _____
Answer _____

3 Arc length



Formula _____
Substitute _____
Answer _____

4 Perimeter



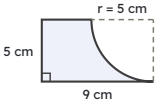
Formula _____
Substitute _____
Answer _____

5 Perimeter



Formula _____
Substitute _____
Answer _____

6 Perimeter

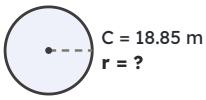


Formula _____
Substitute _____
Answer _____

7 Write the limits of accuracy of 4.5 m. 10.2

_____

8 Radius



Formula _____
Substitute _____
Answer _____

9 Percentage error of 20 cm, to the nearest cm? 10.3

Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A round garden bed has a diameter of 3.2 m. Edging costs $12 per metre. Find the cost to edge the bed, to the nearest dollar.

Answer _____

2 A bike wheel has a radius of 33 cm. How many full turns does it make to travel 1 km?

Answer _____

3 A running track is a rectangle 84 m long with a semicircle on each end. Each semicircle has a diameter of 64 m. Find one lap, to the nearest metre.

Answer _____